

# Assignment 1: Chatbot Development

## Saraswati College Campus Assistant Chatbot

### Objective

The objective of this chatbot is to help students quickly access information related to Saraswati College of Engineering. The chatbot answers common student queries such as admission details, course information, fees, placement details, college timings, departments, and contact information.

### Target Users

- Current Students
- New Admission Applicants
- Parents
- Visitors seeking college information

### Technologies Used

| Layer    | Technology                         |
|----------|------------------------------------|
| Frontend | HTML, CSS                          |
| Backend  | Python, Flask Framework            |
| ML / NLP | CountVectorizer, Cosine Similarity |
| Dataset  | CSV file with college Q&A pairs    |

### Working of the Chatbot

- Step 1:** User enters a query through the web interface.
- Step 2:** The query is converted into text vectors using CountVectorizer.
- Step 3:** Cosine Similarity is calculated between the user query and stored questions.
- Step 4:** The question with the highest similarity score is selected.
- Step 5:** The corresponding answer is displayed to the user.
- Step 6:** If similarity is below threshold, the chatbot shows a fallback message.

### Conversation Flow

| Welcome Message                                                           |                                                |
|---------------------------------------------------------------------------|------------------------------------------------|
| Bot: "Hello! Welcome to Saraswati College Assistant. How may I help you?" |                                                |
| ↓ User Query                                                              |                                                |
| ↓ Convert Query into Vector                                               |                                                |
| ↓ Calculate Cosine Similarity                                             |                                                |
| ↓ Best Match Found?                                                       |                                                |
| YES → Return Answer                                                       | NO → "Sorry, I can't help with this question." |

## Sample Interactions

The following screenshots show actual conversations with the SCOE Campus Assistant Chatbot.

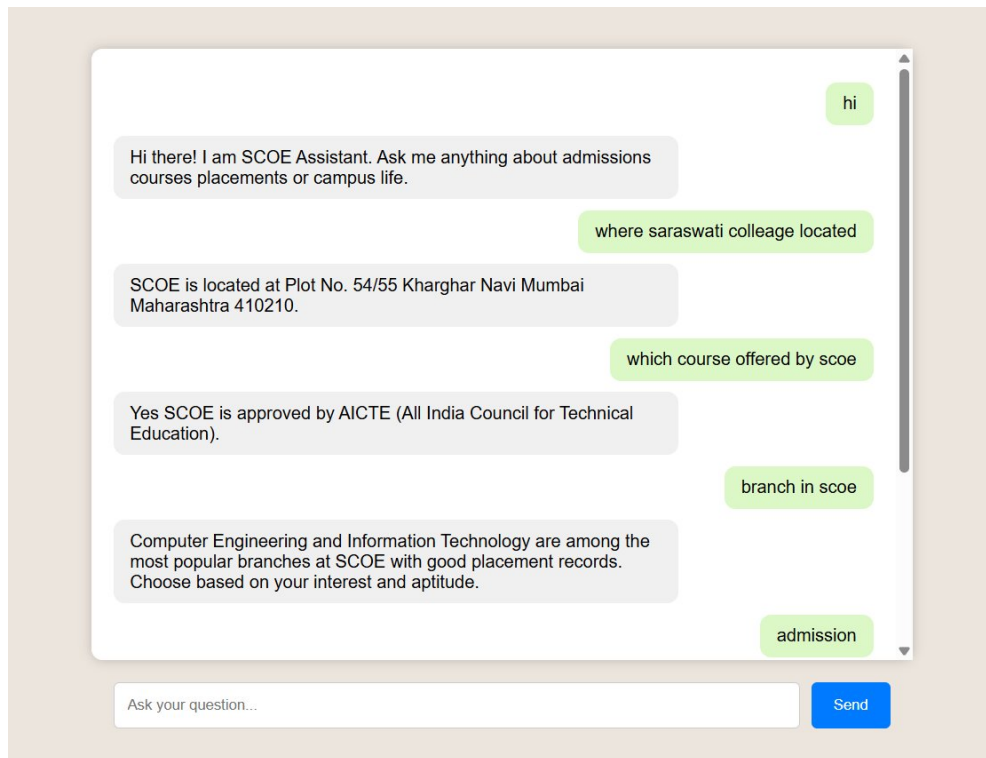


Fig 1: Initial conversation – location, courses, branch queries

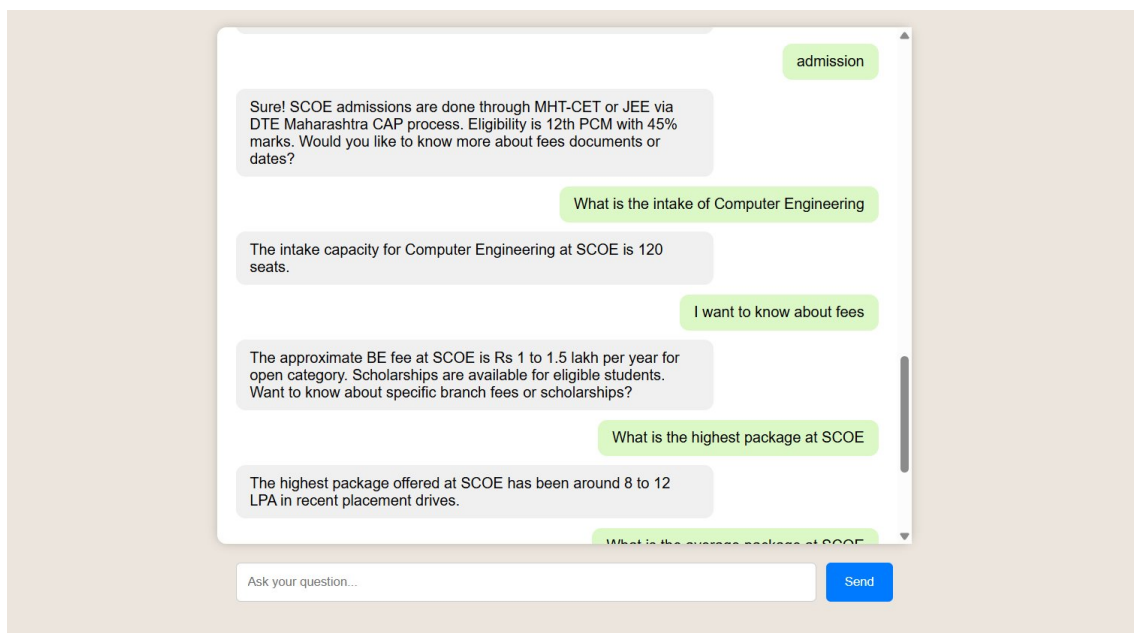


Fig 2: Admission process, fees, and placement package queries

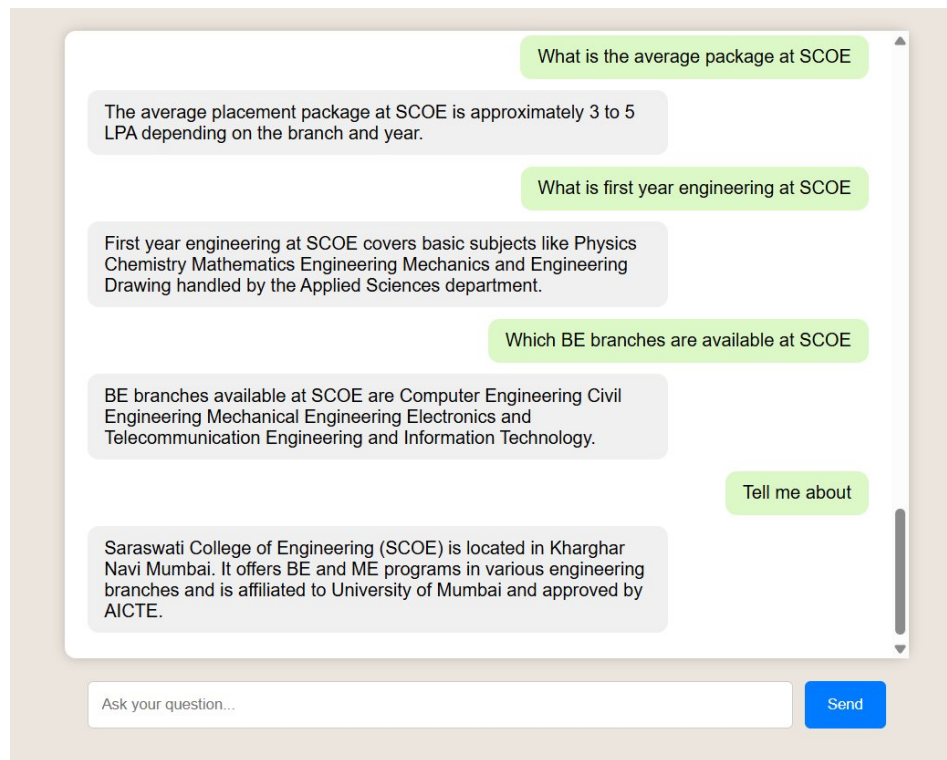


Fig 3: Continued conversation – average package, first year details

## Fallback Response

If the chatbot cannot understand or find a matching question, it displays: **"Sorry, I can't help you with this question."**

## Features

- User-friendly web interface
- NLP-based question matching using Cosine Similarity
- Real-time responses
- Fallback handling for unrecognized queries
- Easy to update dataset using CSV file
- Supports multiple college-related query categories

## Conclusion

The Saraswati College Campus Assistant Chatbot provides students and visitors with instant access to important college information. It reduces manual effort and improves the accessibility of information through an interactive web-based chatbot system built using Python, Flask, and NLP techniques.